

Exploring the psychological mechanisms of metaverse game experiences: Identity construction and emotional impact

HanXi Li ^{a,b}, YuanXiong Liu ^{a,b}, Younghwan Pan ^{b,*}

^a Smart Experience Design Department, Graduate School of Techno Design, Kookmin University, 77 Jeongneung-Ro, Seongbuk-Gu, Seoul, South Korea

^b China-Korea International Institute of Visual Arts Research, Qingdao University of Science and Technology, No. 99 Songling Road, Laoshan District, Qingdao, Shandong Province, China

ARTICLE INFO

Keywords:

Metaverse games
Experience economy theory
Virtual self-identity
Self-estrangement
PLS-SEM

ABSTRACT

Metaverse games offer immersive environments where users construct and engage with idealized virtual identities. However, limited research has explored the psychological mechanisms of virtual self-identity formation and its emotional impact. This study integrates Experience Economy Theory (EET), Self-Identity Theory, and Self-Discrepancy Theory to examine how metaverse gaming experiences influence users' psychological states. Using survey data from 383 metaverse games users and PLS-SEM analysis, results confirm that all four EET dimensions positively impact virtual self-identity, with entertainment experience having the strongest effect and escapist experience the weakest. Virtual self-identity exhibits dual effects, increasing anxiety while lowering self-esteem, with self-estrangement playing a key mediating role in amplifying these negative psychological outcomes. This study extends EET to metaverse gaming, empirically validates the self-identity and self-discrepancy processes in virtual environments, and highlights the psychological risks and benefits of virtual self-identity. Findings offer insights for game designers, users, and policymakers to balance engagement with well-being.

1. Introduction

The rise of the metaverse has brought about transformative changes in the gaming industry, positioning games as both adopters and pioneers of this digital revolution. Its rapid development not only signifies a major shift in digital interaction but also plays a crucial role in driving the multidimensional expansion of the metaverse [1]. The metaverse and gaming are mutually reinforcing, collectively shaping a new digital cultural ecosystem [2]. Metaverse games not only exemplify the potential of technological innovation but also redefine entertainment and virtual interaction experiences [3]. Compared to traditional games, metaverse games offer a highly immersive, interactive, and co-creative gaming environment, gradually evolving into a cultural phenomenon and an artistic medium. They create a continuous, dynamic, interconnected, and malleable virtual gaming world for users [4,5]. According to market projections, the metaverse gaming industry is expected to grow at a compound annual growth rate (CAGR) of 40.4 % from 2023 to 2030 [6]. This rapid expansion is reshaping the landscape of digital entertainment and social interaction, making it a critical topic in the future of online ecosystems.

In metaverse games, virtual avatars serve as a medium for users to shape and express their ideal selves, granting them greater freedom and engagement while fostering new modes of digital identity and virtual social interaction [7]. As metaverse games become more widespread, their impact extends beyond mere entertainment. Within the gaming experience, users not only gain diverse entertainment stimuli but also experience complex psychological tensions between their virtual and real selves [8]. This psychological influence positions metaverse games as a crucial window for studying the interaction between virtual and real identities.

The core appeal of metaverse games lies in their diverse user experiences. Existing studies have highlighted the critical role of factors such as immersion, entertainment, social functionality, and gamification design in influencing user experience, satisfaction, and continued engagement in metaverse gaming [9–13]. However, most research remains limited to analyzing individual dimensions or specific behavioral factors, lacking a systematic theoretical framework to integrate these elements and reveal their overall impact mechanisms. Experience Economy Theory (EET) places user experience at its core, emphasizing the creation of emotional resonance and psychological influence

* Corresponding author at: Smart Experience Design Department, Graduate School of Techno Design, Kookmin University, 77 Jeongneung-Ro, Seongbuk-Gu, Seoul, South Korea.

E-mail addresses: lhxi678@foxmail.com (H. Li), liuyi42556@gmail.com (Y. Liu), peterpan@kookmin.ac.kr (Y. Pan).

<https://doi.org/10.1016/j.entcom.2025.101058>

Received 14 March 2025; Received in revised form 29 October 2025; Accepted 15 November 2025

Available online 17 November 2025

1875-9521/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

through immersive and multidimensional experiences [14]. This aligns closely with the characteristics of metaverse games, which fulfill users' emotional needs and psychological engagement through virtual environments [15]. Therefore, this study adopts EET as its theoretical foundation, incorporating four dimensions (entertainment experience, educational experience, esthetic experience, and escapist experience) to construct a comprehensive framework. This framework systematically organizes experiential factors in metaverse gaming and explores their overall psychological impact on users.

Metaverse games not only attract players through multidimensional user experiences but also have a significant impact on users' psychological well-being. From a psychological perspective, users shape and express their ideal selves through virtual avatars, fostering a sense of virtual self-identity [16]. The sense of achievement and satisfaction gained through gameplay helps narrow the gap between their virtual ideal self and real self, leading to positive emotional experiences [17]. However, when users exit the gaming environment and return to reality, the separation between their ideal and real selves can become more pronounced, amplifying negative emotions. Existing research has indicated that users who are deeply immersed in their idealized virtual selves over an extended period are more likely to experience depression [18], low self-esteem, and anxiety in real life [19,20].

Existing studies have revealed both the positive and negative psychological impacts of virtual self-identity on the real self in the context of metaverse gaming. However, most research has focused on describing psychological outcomes rather than exploring the underlying mechanisms and internal psychological changes that drive these phenomena. Additionally, there is a lack of systematic research examining the transition from user experience to psychological response. A deeper investigation into how metaverse games trigger psychological changes through user experience is essential not only for designing healthier virtual gaming environments but also for mitigating negative psychological effects and promoting users' mental well-being.

In summary, prior research has primarily focused on the technological aspects of metaverse games, such as system implementation, framework development, and visual presentation. Studies addressing user experience are relatively fragmented, and although some have examined the positive and negative psychological outcomes of metaverse gaming, few have delved into the underlying mechanisms of these changes. Unlike traditional online gaming psychology, which has largely emphasized immersion or social motivation, metaverse games offer new contexts through virtual identity construction, sustained immersion, and cross-scenario interaction. These features make it necessary to distinguish metaverse gaming from conventional studies and to further extend the exploration of its psychological mechanisms.

To address this research gap, this study adopts Experience Economy Theory (EET), Self-Identity Theory, and Self-Discrepancy Theory as its core theoretical foundations to construct a comprehensive framework. This framework consists of three layers: Experience Layer (encompassing entertainment experience, educational experience, esthetic experience, and escapist experience); Internalization Layer (focusing on virtual self-identity); Response Layer (including anxiety, self-estrangement, and self-esteem). The study aims to systematically examine experiential factors in metaverse gaming, explore the process of virtual self-identity construction, and investigate its impact on users' mental well-being. By applying this structured framework, the research not only deepens the understanding of psychological mechanisms in metaverse games users but also provides theoretical and practical guidance for designing healthier and more positive virtual gaming environments. Ultimately, this study contributes valuable insights to user experience research and social psychological interventions. This study addresses the following research questions:

1. How do experiential factors in metaverse games influence users' virtual self-identity?

2. What is the role of virtual self-identity in the relationship between metaverse gaming experiences and users' psychological changes?
3. What are the transitional dynamics and psychological mechanisms between users' virtual and real selves?
4. Do different psychological states interact with each other? If so, what are the underlying internal mechanisms?

2. Theoretical background

2.1. Metaverse games

The emergence of the metaverse has brought transformative changes to the gaming industry, attracting widespread global attention and participation [21]. It not only marks a significant shift in digital interaction but also has the potential to redefine the dynamic relationships between users, technology, and virtual society [22,23]. Prior studies generally regard gaming platforms that feature virtual identity construction, immersive environments, user-generated content, and sustained social interaction as metaverse games [1,10,24–26]. Such games go beyond the traditional paradigm of entertainment by offering immersive and interactive experiences that blur the boundaries between the physical and virtual worlds, thus creating a more open and continuous digital space [10]. Core characteristics of metaverse games include social interaction, interactivity, immersion, and user-generated content [3,27,28]. Within these socially connected and interactive virtual environments, users are able to cultivate and sustain meaningful friendships [21].

Metaverse games, represented by Animal Crossing, Minecraft, Roblox, Play Together, and Eggy Party, have evolved from simple entertainment platforms into virtual spaces with significant economic, cultural, and social impact [2,29]. Since its release, Animal Crossing has deeply integrated into users' social and daily lives, becoming a popular venue for virtual gatherings [30]. Minecraft has sold 300 million copies and reached 150 million monthly active users as of October 2023, generating substantial revenue through its in-game marketplace [31]. Roblox has surpassed other social networking applications in terms of total monthly usage time [32]. Play Together attracted over 100 million downloads within a year, appealing to a broad audience of casual and sandbox game enthusiasts [24]. Meanwhile, Eggy Party recorded 47.33 million daily active users in December 2022, making it NetEase's highest-grossing game and its most-played title in terms of daily active users [33]. The widespread success of these metaverse gaming platforms reflects users' growing demand for innovative entertainment experiences, driving rapid market growth and highlighting the potential of metaverse games as digital cultural platforms [34].

Existing research has explored various aspects of how metaverse games influence user experience. Jo and Lee [10] analyzed the impact of presence and seamlessness in metaverse games, emphasizing their critical role in enhancing user immersion, escapism, and self-expression. Study [21] examined determinants of user experience in metaverse games, finding that improving entertainment value and immersion, while simplifying interfaces and optimizing social functions, is key to enhancing user experience. Thomas [13] demonstrated that nudge experience, flow experience, alternate reality experience, and hedonic experience in metaverse games enhance user engagement and immersion, enriching the gamification experience. Cheremnykh [9] conducted a case study on Fortnite, analyzing how gamification design, social interactions, and user customization shape the user experience, while highlighting the importance of balancing user freedom with system structure. While these studies have identified key factors affecting user experience in metaverse games, a comprehensive framework is still lacking to systematically integrate these experiential elements and explain their overall impact mechanisms.

The depth of user experience in metaverse games can also lead to complex psychological changes. Jo [26] suggested that metaverse platform design should balance users' psychological needs by

integrating both free exploration and structured system guidance. Taylor [35] further emphasized that the immersiveness, interactivity, and virtual reality features of the metaverse have a direct impact on users' psychological states, influencing self-actualization and personal growth over prolonged use. However, the psychological effects of metaverse games are not always positive. Research [36] found that metaverse games can exacerbate mental health issues, with users who engage in Zepeto as an escape from reality being more likely to experience loneliness and a lack of social support. Suh [37] described the contradictory emotional experiences users encounter in virtual worlds, such as excitement and anticipation from social interactions, alongside anxiety and depression from detachment from real life. Zumrut [38] explored the metaverse's impact on self-perception, self-esteem, and the balance between virtual and real life. Similarly, Usmani [39] highlighted that excessive time spent in metaverse gaming can lead to insecurity, anxiety, depression, and even behavioral addiction. These studies indicate that metaverse games not only reshape user behaviors and emotional patterns but also pose potential mental health risks over prolonged use. Therefore, to comprehensively understand users' psychological changes in virtual worlds, this study further explores the potential mechanisms underlying the psychological impact of metaverse gaming experiences. The findings aim to provide psychological insights for future game design, fostering healthier and more balanced virtual gaming environments.

In summary, existing research has revealed the dual impact of metaverse games on user behavior and psychology: offering positive experiences such as immersion and self-expression, while also posing potential mental health risks. Building on these insights, this study will further explore the complex experiential mechanisms in metaverse gaming and their psychological effects on users. The goal is to provide a deeper understanding of these interactions and offer guidance for designing healthier, more enjoyable, and balanced gaming experiences.

2.2. Experience Economy Theory (EET)

Pine and Gilmore [40] established a clear economic distinction between goods, services, and experiences, introducing the concept of the Experience Economy. They emphasized that economic structures are shifting from a goods-based and service-based economy toward an experience-driven economy [41]. This shift suggests that businesses should go beyond merely providing products and services; instead, they should design and deliver meaningful, valuable, and unique experiences. EET consists of two fundamental dimensions: participation and connection. Participation refers to the user's role in the experience, which can range from passively receiving to actively engaging. Connection describes the relationship between the user and the experience, varying from absorbing the content to fully immersing in the experience environment. Based on these two dimensions, Pine and Gilmore defined four distinct types of experiences (Fig. 1): When users passively participate but focus on the experience content, they develop an entertainment experience; when they actively participate and absorb knowledge, they form an educational experience; when they immerse themselves in the environment without actively engaging, they experience an esthetic experience; and when they actively participate and fully integrate into the experience environment, they undergo an escapist experience.

EET has been widely applied in various fields related to user experience research, with tourism being the most studied area. Previous research has confirmed that the four experiential dimensions of EET significantly influence tourist experiences, which in turn affect destination word-of-mouth, loyalty, and satisfaction [14,42–44]. Additionally, EET has been applied to VR experiences [45], AR experiences [46], the cruise industry [47], and festival marketing [48], among other relevant fields. However, EET has not yet been applied to experience research in the metaverse domain. Previous studies indicate that EET primarily focuses on user experience, aiming to meet user needs, create

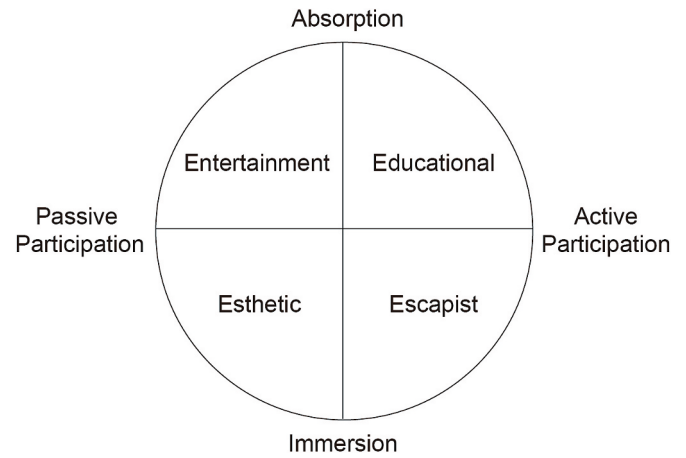


Fig. 1. The four dimensions of experience.

memorable experiences, evoke positive emotions, and stimulate deep psychological and behavioral responses [49]. Given that metaverse games are inherently immersive and interactive experiential forms, they provide fully immersive virtual environments that enable users to experience highly engaging and multisensory interactions [26]. These games closely connect users' emotions and psychological states with virtual environments, allowing for profound psychological and emotional experiences, making them highly aligned with EET's research focus.

Specifically, the four experiential dimensions of EET are fully reflected in metaverse games. For instance, rich interactivity and engaging elements can attract users to participate and focus on game content, thereby creating an enjoyable entertainment experience [34]. Through gamified learning, completing virtual tasks, or participating in challenges, users can acquire knowledge and develop skills, leading to an educational experience [50]. The design of esthetic elements, such as virtual game environments, interactive dynamics, and visual effects, enhances the game's appeal and immersion, contributing to an esthetic experience [51]. Additionally, metaverse games provide users with a virtual space detached from reality, allowing them to immerse themselves, relieve stress, and enjoy a temporary escapist experience [8].

Existing research has identified the metaverse as one of the best examples of the Experience Economy [15]. Applying EET to metaverse games research not only provides a systematic theoretical framework for understanding metaverse experiences but also extends the experience concepts of EET from the physical world to the virtual realm. This extension enables a deeper understanding and explanation of users' psychological and behavioral responses in metaverse gaming environments.

2.3. Self-identity theory

Self-Identity Theory describes how individuals understand and define themselves through self-concept, exploring their roles, sense of belonging, and self-identity within social contexts. These factors shape individual behavior and perceptions of self-worth [52]. Self-identity represents how individuals define themselves and is influenced by motivation, social interactions, and role-playing [53], playing a crucial role in motivating behavior. Specifically, individuals compare others' expectations with their own values, beliefs, and experiences, and through internalization, they transform these external influences into personal self-expectations [54].

In metaverse games, Self-identity Theory is further expanded. Users extend their self-concept into the digital environment through virtual avatars, forming a virtual self [7,17]. They can also enhance their virtual self through virtual characters and digital possessions [55,56]. The virtual self serves as a platform for users to explore and express their

ideal self [57], allowing them to realize their aspirations and visions within the virtual world. Consequently, the concept of self-identity extends into the virtual realm as virtual self-identity, where users express or align their virtual avatars with their ideal self, thereby experiencing a reinforced sense of self-identity within the metaverse [8].

For example, in metaverse games, users may choose or design a more attractive, powerful, or confident avatar [58] to embody traits they lack in reality but aspire to possess. These virtual avatars align with their ideal self-characteristics, offering users distinct cognitive and psychological experiences that differ from those in the real world, thereby enhancing their self-identity within the metaverse. Consequently, virtual self-identity serves as an extension or representation of users' ideal self, allowing them to resonate with their virtual character experiences [19].

Furthermore, the Proteus Effect [59] further enriches the understanding of virtual self-identity, suggesting that users' behaviors and psychological states may be influenced by the characteristics of their virtual avatars [19]. This effect provides insights into the dynamic nature of virtual self-identity in metaverse games, explaining how virtual character shape users' psychological states and influence their behaviors within the virtual environment.

In metaverse games, the transition between virtual self and real self is not a simple shift but rather a dynamic process involving cognitive reconstruction, emotional regulation, and social role perception. Studies have shown that virtual self-identity has profound effects on users' behavioral motivation in virtual games [56,60], self-perception [61,62], and mental health [7,16,18,63]. The stronger a user's identification with their virtual avatar, the greater their emotional investment and psychological attachment, which can, in turn, increase the risk of problematic gaming behaviour [8].

Although identity theory emphasizes that the formation of virtual self-identity is a dynamic, complex, and multidimensional process involving the interaction of multiple identity facets and their evolution over time [64,65], in the field of game studies scholars have commonly operationalized this construct through users' attachment to and projection onto their virtual characters [60,61,66]. This approach effectively captures the core aspects of identity experience in gaming and has been widely applied to explain psychological mechanisms in virtual environments. Building on this tradition, the present study conceptualizes virtual self-identity as users' emotional attachment to and identity extension through their virtual characters, in order to better capture the underlying psychological mechanisms in empirical analysis.

2.4. Self-discrepancy theory

Self-Discrepancy Theory, proposed by E. Tory Higgins [67], defines three self-states: the Actual Self, which represents an individual's perceived real attributes; the Ideal Self, which reflects the traits an individual aspires to have; and the Ought Self, which encompasses the traits an individual believes they should possess, often influenced by social responsibilities and external expectations [67]. Self-Discrepancy Theory explains how discrepancies between these self-states impact emotions and psychological well-being. When there is a significant gap between an individual's Actual Self and Ideal Self or Ought Self, they are more likely to experience negative emotions such as disappointment, anxiety, depression, dissatisfaction, and fear. The greater the self-discrepancy, the stronger the negative emotional responses [68].

With the rise of virtual environments, many studies have extended Self-Discrepancy Theory into the digital realm [61], defining the gap between actual self-identity and ideal self-identity in virtual environments as Virtual Self-Discrepancy [69]. VSD has been recognized as a key factor in explaining problematic behaviors in virtual worlds [70] and plays a crucial role in understanding users' psychological perceptions of their virtual selves as well as their interpersonal responses in social interactions [71]. When a user's virtual self closely aligns with their ideal self, they may experience enhanced confidence and self-

esteem. Conversely, if there is a large discrepancy between the virtual and real self, users may face psychological setbacks upon returning to reality, leading to emotional fluctuations and cognitive dissonance.

Research widely agrees that in virtual environments, avatars serve as idealized versions of players' actual selves [61,71–73]. Similarly, in metaverse games, users shape and express their ideal selves through avatars, often embodying traits that are difficult to achieve in real life [72,74]. When users experience a sense of achievement and satisfaction in metaverse games, their virtual self-discrepancy temporarily narrows. However, upon exiting the game and returning to reality, the separation between the ideal self and actual self becomes more pronounced, causing a sudden amplification of VSD and triggering negative emotions [70]. For instance, studies have found that users deeply immersed in idealized virtual selves are more likely to experience depression [72], low self-esteem [75,76], and anxiety in real life [20,77]. These findings confirm that Self-Discrepancy Theory plays a crucial role in explaining the psychological impact of virtual gaming on users' mental health.

Although existing studies have revealed both positive and negative psychological effects of virtual self-identity, most of them focus on descriptive analyses of outcomes, lacking in-depth exploration of the dynamic interaction between virtual and real selves and the underlying psychological mechanisms. Metaverse games provide users with highly immersive platforms for identity expression and ideal-self construction; however, it remains unclear how the psychological gap between virtual and real selves triggers emotional fluctuations when users return to the real world. To address this issue, the present study draws on Self-Discrepancy Theory to explain the psychological consequences of discrepancies between virtual and real selves. Building on these insights, this study extends the scope of discrepancy outcomes to include self-estrangement, which reflects a broader form of detachment from the real self. On this basis, we introduce self-estrangement as a key mediating variable in our model, thereby linking theoretical insights with empirical analysis.

2.5. Summary

In summary, these theoretical frameworks jointly provide a foundation for understanding user behavior and psychological changes in the metaverse gaming environment. Specifically, EET explains users' experiences in metaverse games through four dimensions: entertainment, education, esthetic, and escape. Self-identity Theory and Self-Discrepancy Theory further explain how, during these experiences, users construct their ideal selves through virtual avatars, thereby reinforcing virtual self-identity. By integrating these perspectives, the present study not only examines the relationship between experiential factors and psychological well-being but also uncovers a two-step mediating pathway: first, virtual self-identity serves as a key mediator between user experiences and psychological outcomes; subsequently, self-estrangement functions as a deeper psychological mechanism that explains how virtual self-identity translates into outcomes such as anxiety and self-esteem. The proposed research model is presented in Fig. 2.

3. Hypothesis development

3.1. Experience layer – internalization layer

3.1.1. Entertainment experience

Entertainment experience refers to the sense of joy and relaxation that users derive from engaging in fun and enjoyable activities [45]. Research suggests that entertainment experiences not only help users relieve stress and fatigue but also enhance overall well-being and psychological comfort [78]. In metaverse games, entertainment experience is manifested through engaging and enjoyable challenges, social interactions with other players, and creatively designed game environments, all of which contribute to a sense of pleasure and satisfaction [34].

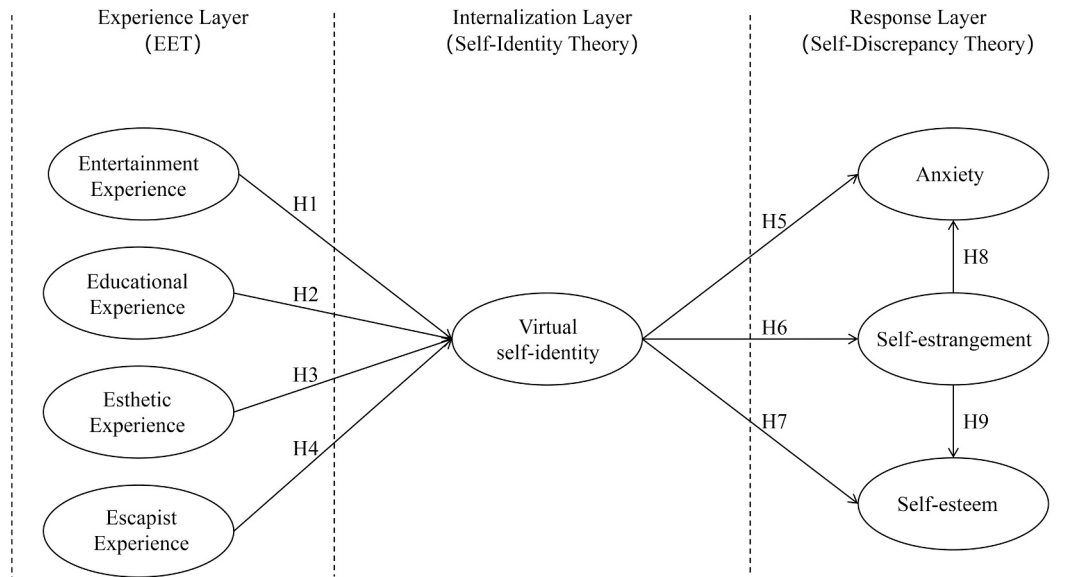


Fig. 2. Research model diagram.

Previous research has indicated that entertainment experiences provide users with positive and enjoyable emotional states [79], encouraging deeper engagement in role-playing and social interactions within gaming contexts [56]. This immersive entertainment process strengthens the emotional bond between users and their virtual characters, thereby enhancing their identification with the virtual self [80].

Therefore, the gamified elements of metaverse games, including social interactions, emotional connections with other virtual characters, and creative freedom in gameplay, are proposed to enhance users' identification and sense of belonging toward their virtual avatars. Based on this, the study proposes the following hypothesis:

H1: Entertainment experience has a positive effect on users' virtual self-identity.

3.1.2. Educational experience

Educational engagement refers to the process of capturing users' attention by teaching new skills, providing knowledge, or presenting complex problem-solving tasks, which stimulates interest and a desire for learning [47]. Educational experience is an active process of participation and absorption, which not only enhances users' cognitive abilities but also fosters a sense of achievement and psychological satisfaction [46,49]. Previous studies on tourism experiences, VR museums, and VR concerts have confirmed that educational experiences have significant positive effects on users' psychological states and behaviors [44,45,81].

In metaverse games, educational experience refers to the process in which users gain knowledge and enhance skills by completing challenging tasks, solving puzzles, or engaging in strategic interactions. These tasks are often designed with a combination of interactivity and entertainment, allowing users to not only acquire new knowledge but also experience a sense of achievement and satisfaction through problem-solving. For example, in the *Eggy Party* and *Detective Conan* crossover game, users are required to search for clues and apply reasoning strategies to progress through the game. By completing these tasks, users not only develop their analytical and problem-solving skills but also experience a strong sense of pride and fulfillment from successfully solving mysteries.

It can be observed that educational experiences create a positive, safe, yet challenging gaming environment that enables users to continuously expand their abilities when facing challenges. This process enhances users' confidence in their own capabilities, known as self-efficacy, and gradually internalizes into a positive self-identity [82].

Furthermore, real-time feedback and achievement mechanisms in games further reinforce users' confidence in their abilities and strengthen their sense of belonging to their virtual characters. Based on this, the study proposes the following hypothesis:

H2: Educational experience has a positive effect on users' virtual self-identity.

3.1.3. Esthetic experience

Esthetic quality refers to the visual and design elements conveyed through color, imagery, typography, and layout [45]. Esthetic design plays a crucial role in user experience, particularly in highly immersive virtual environments, where its impact becomes even more pronounced [83]. Research suggests that esthetic appeal not only compensates for technological limitations [83] but also elicits users' esthetic emotions, thereby enhancing their satisfaction with virtual environments [81]. In this context, esthetic design is not merely a functional supplement but a key driving force in shaping users' psychological states and behaviors.

In metaverse games, esthetic experience is reflected in the richly designed environments, character appearances, and virtual architecture, all of which provide players with highly refined visual effects and strong visual enjoyment. For example, in *Fortnite*, the gradient sky colors, unique building textures, and diverse character skins create a vivid and captivating visual experience. These design elements not only meet users' visual expectations but also enhance the overall appeal of the game.

Esthetic experience is not merely passive visual enjoyment; it also serves as a platform for users to express their personality and values. In the virtual world, users can customize their character's appearance, select unique decorations, or design personalized structures, showcasing their esthetic preferences. This esthetic expression allows users to reinforce their ideal self-image in the virtual world, thereby enhancing their identification and sense of belonging to their virtual avatars. Research suggests that when users are satisfied with their virtual character's appearance, they are more likely to internalize the avatar's idealized traits as part of their virtual self-identity [84]. Based on this, the study proposes the following hypothesis:

H3: Esthetic experience has a positive effect on users' virtual self-identity.

3.1.4. Escapist experience

In EET, escapist experience refers to users temporarily distancing themselves from real-world stress and concerns, achieving psychological

relief and comfort through immersive experiences, which helps generate positive emotions [46]. Research suggests that gaming serves as a means of escaping reality and detaching from everyday life [85]. In metaverse games, escapism manifests in users' immersion in surreal virtual environments, where they can enjoy freedom and exploration while temporarily forgetting real-world worries. For example, in Minecraft, users can freely construct their ideal world, designing houses, cities, and even reshaping natural landscapes, experiencing a sense of absolute control and creative freedom.

Studies indicate that escapist behavior occurs across different game genres through virtual identity identification, where stronger escapist motivation leads to stronger identification with one's virtual avatar [8]. Moreover, escapism has been found to predict and mediate the relationship between gaming behavior and users' negative psychological outcomes [86]. Additionally, escapist experience are closely linked to flow states—as users immerse themselves in their virtual characters, they are more likely to enter a state of flow, leading to a deeper attachment to their virtual identity and environment [87], ultimately strengthening virtual self-identity. Based on this, the study proposes the following hypothesis:

H4: Escapist experience has a positive effect on users' virtual self-identity.

3.2. Internalization layer – response layer

According to Self-identity Theory, virtual self-identity is an internalized psychological process, representing the alignment between one's virtual avatar and ideal self. Users express their ideal self through their virtual avatars, fulfilling their desire for identity construction [7,17]. However, based on Self-Discrepancy Theory, when there is a significant gap between the ideal self and the real self, users experience a psychological detachment state, manifesting as a misalignment between the ideal self and actual self [69]. In virtual environments, users derive satisfaction from their idealized virtual self, but when this virtual identity is brought back into reality, they may experience frustration, contradiction, or identity confusion due to the limitations of their real self [7]. Specifically, this psychological transition mechanism can be understood as follows: virtual self-identity enhances users' perception of their ideal self, but when the real self fails to match this perception, it triggers psychological conflict. This mechanism forms the theoretical foundation for understanding how virtual self-identity influences users' psychological states.

This psychological transformation mechanism has been studied across various fields. For instance, in virtual communities, research has found that users' identification with their virtual avatar and the virtual community significantly influences self-efficacy, collective efficacy, and trust, which in turn affects their willingness to continue using the virtual community [17]. In the social media domain, users engage in self-promotion and identity expansion by curating an idealized virtual self, and this virtual self-identity has been shown to significantly impact users' behavioral intentions and real-world actions [88]. Additionally, the construction of virtual self-representations has been found to affect emotions and self-esteem in young female users. Prolonged exposure to idealized self-images on social media can lead to aesthetic pressure and emotional fluctuations related to dissatisfaction [89]. A study on Gen Z Facebook users in Egypt further demonstrated that a false self-representation significantly increases social anxiety [63]. In the gaming domain, virtual self-identity has also been proven to have profound effects on psychological states. In video games, research suggests that Player Avatar Identification (PAI) facilitates self-extension, where players with a dispersed identity style tend to score higher in focus and identity recognition [66]. In the context of massive multiplayer online role-playing games, players engage in Avatar Identification and Avatar Embodiment, perceiving their virtual characters as extensions of themselves, which modifies their attitudes and behaviors while establishing psychological connections and emotional bonds with their avatars. This

process significantly enhances social motivation, immersion motivation, and achievement motivation, ultimately increasing users' engagement and willingness to participate in the game [12].

Furthermore, the psychological transformation mechanism has been widely recognized for its profound impact on users' mental and emotional states. Research indicates that virtual self-identity may contribute to various real-world psychological issues and emotional disorders [90]. For instance, multiple studies have found that the stronger the users' psychological attachment to their virtual avatars, the higher the risk of problematic gaming behaviors. Users' psychological association with their game characters can lead to temporary shifts in self-perception [91], which may trigger anxiety [16], feelings of loss [7], social anxiety [16], and depression [92]. It can also reinforce escapism and avoidance coping behaviors [8], loneliness [93], and impulsivity [94]. Additionally, stronger virtual self-identity is associated with lower real-world self-esteem [95].

Metaverse games go beyond the boundaries of traditional gaming, exhibiting high levels of immersion, social connectivity, and co-creativity [3,28]. These characteristics make users more likely to perceive their virtual avatars as projections of their ideal selves, reinforcing virtual self-identity and further intensifying the disconnection between the ideal self and actual self. Specifically, on one hand, the immersive experience of metaverse games makes it easier for users to fully engage with their virtual avatars. On the other hand, the limitations of the real self in everyday life make it difficult to sustain an idealized virtual identity, which can trigger psychological reactions similar to those found in previous studies. Based on this, the study proposes the following hypotheses:

H5: Virtual self-identity has a positive effect on anxiety.

H7: Virtual self-identity has a negative effect on self-esteem.

Self-Alienation is a significant theoretical concept in sociology and psychology, systematized and widely disseminated by Seeman. He categorized alienation theory into five dimensions: powerlessness, meaninglessness, normlessness, isolation, and self-estrangement [96]. Among these, self-estrangement is a core dimension of alienation theory, referring to an individual's sense of detachment from oneself, often manifested as an inability to align with core personal needs or a loss of identification with the true self. This disconnection is reflected in emotional, behavioral, or role-related inconsistencies [96,97].

In virtual environments, self-estrangement can be understood as a phenomenon in which individuals, upon strongly identifying with their idealized virtual avatars, experience a blurring of the boundaries between their real and virtual identities [98]. When users return to reality, the limitations of their real selves can create internal psychological conflict [99]. Specifically, when users engage in metaverse gaming, they may experience a sense of fulfillment by expressing aspects of their ideal self that they cannot achieve in real life. However, this satisfaction is often temporary, once users become aware of the mismatch between their real self and their virtual avatar, they may experience psychological conflict and a sense of detachment from their true self. Studies have pointed out that virtual self-identity plays a crucial role in this process. By fostering deep engagement with and recognition of their virtual avatars, virtual self-identity enhances users' perception of their ideal self. At the same time, however, it also intensifies the gap between the virtual and real self, ultimately leading to self-estrangement [99,100].

Self-estrangement manifests in individuals through emotional and behavioral changes by weakening self-identity and intrinsic motivation, which indirectly affects their engagement in tasks and sense of accomplishment [101]. On one hand, self-estrangement amplifies the inconsistency between the virtual and real self, increasing emotional distress. Studies suggest that when users realize that their virtual avatar's performance does not align with their real-life self, they may experience unease, tension, and anxiety [102,103]. On the other hand, when individuals rely heavily on virtual avatars for self-expression, their real self's sense of worth and presence may be diminished, leading to self-estrangement. Research has found that higher levels of self-

estrangement weaken an individual's sense of self-worth, and when self-estrangement intensifies, self-esteem tends to decline [101]. Therefore, self-estrangement may serve as a mediator between virtual self-identity and psychological states (anxiety and self-esteem). Based on the above analysis, the following hypotheses are proposed:

H6: Virtual self-identity has a positive effect on self-estrangement.

H8: Virtual self-identity positively influences anxiety through self-estrangement.

H9: Virtual self-identity negatively influences self-esteem through self-estrangement.

4. Methodology

4.1. Participants and data collection

This study employed a survey method to collect data. To enhance sample diversity, convenience sampling was conducted by openly recruiting participants through Chinese social media platforms such as Xiaohongshu, Zhihu, Weibo, and WeChat. This approach enabled access to a naturally available population and facilitated the rapid acquisition of a large number of respondents. At the same time, purposive sampling was applied by setting clear eligibility criteria: only users who had prior experience with metaverse games were invited to participate. This ensured that respondents had sufficient familiarity with the platforms and relevant gameplay experience, thereby aligning the sample with the research objectives. The survey covered widely recognized metaverse games cases, including Roblox, Animal Crossing, Minecraft, Fortnite, Eggy Party, Grand Theft Auto, Axie Infinity, and Second Life, all of which have large user bases and are representative in this domain [10]. The sampling strategy also sought to include participants across different age groups, educational backgrounds, and regions in order to enhance diversity and, to some extent, improve the representativeness of the results.

To mitigate potential non-response bias [104], several measures were implemented. First, multiple rounds of survey invitations were issued to broaden user coverage and ensure that participants with different backgrounds and characteristics had access to the questionnaire. Second, the invitation process emphasized the academic purpose and anonymity of the study, thereby alleviating concerns about privacy disclosure and improving response rates. In addition, to reduce the risk of self-selection bias, the invitation explicitly noted that the opinions of users with lower levels of technological interest or less frequent engagement with metaverse games were equally valuable, thereby encouraging a broader range of users to participate. Through these measures, the study sought to maximize sample diversity and data quality within the constraints of accessible conditions.

Given that this study involves potentially sensitive data related to users' virtual avatars, behavioral experiences, and psychological perceptions, all data collection and processing are conducted anonymously, strictly adhering to ethical guidelines to protect user privacy. The questionnaire consists of four sections: the first section of the questionnaire asks users whether they have experienced metaverse games such as Roblox, Animal Crossing, Minecraft, Fortnite, Eggy Party, Grand Theft Auto, Axie Infinity, and Second Life. If they select "No," the system will automatically terminate the questionnaire; if they select "Yes," they must specify which platforms they have used, ensuring the reliability of the data. The second section introduces the purpose of the study and presents the informed consent form. The third section collects demographic information, including gender, age, education level, usage frequency, and session duration. The fourth section consists of the measurement scales, which are adapted from previous studies with slight modifications to fit the research context. Specifically, the EET measurement scale consists of 16 items, adapted from [14,45]; the self-identity measurement scale consists of 4 items, adapted from [64,66,105]; the anxiety measurement scale consists of 4 items, adapted from [87,106]; the self-estrangement measurement scale consists of 5

items, adapted from [99]; and the self-esteem measurement scale is based on the Rosenberg Self-esteem Scale [107], which consists of 10 items. In total, the questionnaire contains 39 items across 8 measurement variables, with all items using a 5-point Likert scale, where 1 represents "Strongly Disagree" and 5 represents "Strongly Agree." The questionnaire is distributed through Wenjuanxing, allowing users to scan a QR code to participate in the survey.

According to the Structural Equation Modeling (SEM) sample size calculator [108], the minimum required sample size for this study was determined to be 177, based on the model structure and number of variables. A total of 452 responses were collected, and after removing invalid or anomalous responses, 383 valid questionnaires remained. This final sample size exceeds the minimum requirement and meets the recommended sample size for SEM analysis [109].

4.2. Common method bias (CMB)

The data for this study were collected through a questionnaire survey. During the data collection process, factors such as the single format of the questionnaire, similar measurement environments, and individual emotional states could introduce potential biases in the results. To assess common method bias, Harman's single-factor test was conducted [110]. The results indicated that the first unrotated factor explained 19.064 % of the total variance, which is below the 40 % threshold, suggesting that common method bias is not a significant concern in this study. Additionally, multicollinearity diagnostics were performed [111] by analyzing the variance inflation factor (VIF) values of the paths within the structural model. As shown in Table 3, all latent variables VIF values were below 5 [112], indicating that multicollinearity is not an issue and that the constructs exhibit a high degree of independence from one another.

5. Results

5.1. Demographic analysis

Table 1 presents the detailed characteristics of the survey participants. The sample is gender-balanced, with 51.7 % male and 48.3 % female respondents. In terms of age distribution, the majority of participants fall between 18 and 34 years old, accounting for 71.18 % of the total sample, indicating that young users are the primary demographic group engaging with and following metaverse games. Regarding educational background, bachelor's degree holders make up 52.74 % of the sample, suggesting that higher-educated individuals may have a greater interest and engagement in metaverse gaming platforms. In terms of usage frequency, 66.83 % of users play at least once per week, indicating a high level of engagement with metaverse games. For daily playtime, the most common duration is 30 min to 1 h, accounting for 22.45 % of participants, suggesting that most users engage with metaverse games for a moderate duration, aligning with the characteristics of casual gaming. Participants' platform preferences show a diverse range of metaverse games usage. Grand Theft Auto is the most frequently used platform, with 68.93 % of participants reporting engagement. Second Life and Axie Infinity follow closely, with 64.75 % and 58.49 %, respectively. Roblox, Minecraft, and Fortnite have usage rates of 51.44 %, 44.65 %, and 57.96 %, respectively. Since this data is derived from multiple-choice responses, it suggests that many users engage with multiple platforms simultaneously. This result not only reflects users' diverse preferences but also highlights the varying degrees of appeal among different metaverse gaming platforms.

5.2. Reliability, validity, and factor loadings analysis

This study employs Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze the data [113]. PLS-SEM is well-suited for handling complex models with multiple variables—in this case, the

Table 1
Demographic frequency analysis results.

Demographic information	Category	Frequency	Percentage (%)
Gender	Male	198	51.7
	Female	185	48.3
Age	Under 18	10	2.61
	18–24	154	40.21
	25–34	119	31.07
	35–44	34	8.88
	45–54	34	8.88
	Above 55	32	8.36
Education Level	Below Bachelor's	24	6.27
	Bachelor's	202	52.74
	Master's	97	25.33
	Doctorate	60	15.67
Frequency of use	Once per month	42	10.97
	Once every two weeks	57	14.88
	Once per week	103	26.89
	More than once per week	97	25.33
	Once per day	51	13.32
Daily Using Time	More than once per day	33	8.62
	<10 min	7	1.83
	10–30 min	74	19.32
	30 min – 1 h	86	22.45
	1 h–1.5 h	107	27.94
	1.5 h–2 h	72	18.80
	>2 h	37	9.66
Game Title (multiple-choice)	Roblox	197	51.44
	Animal Crossing	116	30.29
	Minecraft	171	44.65
	Fortnite	222	57.96
	Eggy party	66	17.23
	Grand Theft Auto	264	68.93
	Axie Infinity	224	58.49
	Second life	248	64.75

research model includes eight latent variables. This approach emphasizes predictive capability, making it ideal for exploratory research and small sample analysis. Previous studies in online gaming research have also utilized PLS-SEM for analysis [114–116], which aligns well with the objectives of this study. Therefore, SmartPLS 4.0 was chosen as the analytical tool.

Before conducting the analysis, the reliability, validity, and internal consistency of the model were examined, along with factor loadings. Table 2 presents the reliability and validity results of the constructs, including Cronbach's Alpha values, Composite Reliability (rho_a and rho_c), and Average Variance Extracted (AVE). The results indicate that all constructs have Cronbach's Alpha, rho_a, and rho_c values exceeding 0.7, suggesting good reliability and validity. Additionally, AVE values exceed 0.5, confirming satisfactory internal consistency of the questionnaire.

Table 3 presents the factor loadings measurement results. The closer the factor loading coefficient is to 1, the stronger the explanatory power of the observed variable in representing its latent construct [117]. In this study, the factor loadings for all constructs range from 0.794 to 0.901, exceeding the commonly accepted threshold of 0.7. This indicates that the measurement variables effectively represent their respective latent

Table 2
Reliability and validity indicators.

	Cronbach Alpha	Rho_A	CR	AVE
Educational experience	0.867	0.869	0.909	0.714
Entertainment experience	0.846	0.846	0.896	0.684
Escapist experience	0.865	0.875	0.907	0.710
Esthetic experience	0.844	0.884	0.892	0.674
Virtual self-identity	0.836	0.840	0.890	0.670
Self-estrangement	0.878	0.879	0.911	0.673
Anxiety	0.917	0.919	0.942	0.801
Self-esteem	0.957	0.959	0.962	0.719

Table 3
VIF and factor loadings.

	Item	VIF	Factor loading
Anxiety	AN1	3.082	0.901
	AN2	2.844	0.887
	AN3	2.991	0.898
	AN4	2.891	0.895
Escapist experience	EC1	2.118	0.852
	EC2	1.961	0.854
	EC3	2.052	0.826
	EC4	2.036	0.838
Educational experience	ED1	2.078	0.853
	ED2	2.075	0.846
	ED3	2.169	0.841
	ED4	2.002	0.841
Entertainment experience	EN1	2.117	0.851
	EN2	1.794	0.811
	EN3	2.033	0.838
	EN4	1.714	0.807
Esthetic experience	ES1	1.936	0.815
	ES2	1.982	0.855
	ES3	1.828	0.759
	ES4	1.790	0.852
Self-estrangement	SET1	2.252	0.846
	SET2	2.048	0.811
	SET3	1.866	0.794
	SET4	2.293	0.844
	SET5	1.933	0.805
Self-esteem	SEM1	3.443	0.871
	SEM2	2.793	0.833
	SEM3	2.409	0.813
	SEM4	3.413	0.868
	SEM5	3.121	0.851
	SEM6	2.910	0.846
	SEM7	2.686	0.825
	SEM8	2.956	0.845
	SEM9	3.269	0.866
	SEM10	3.149	0.860
Virtual self-identity	VSI1	1.809	0.810
	VSI2	1.815	0.821
	VSI3	1.811	0.834
	VSI4	1.790	0.807

constructs, ensuring high explanatory power and validity of the measurement indicators.

5.3. Discriminant validity

To assess discriminant validity, this study evaluates whether the constructs have sufficient independence from one another. Two commonly used methods for this verification are the Fornell-Larcker criterion [118] and the Heterotrait-Monotrait (HTMT) ratio [119]. According to the results presented in Table 4, the square root of the AVE (diagonal values) for each construct is greater than its correlations with other constructs, indicating good discriminant validity between variables based on the Fornell-Larcker criterion.

Additionally, Table 5 presents the HTMT ratio results, which further confirm discriminant validity. All HTMT values are well below the 0.85 threshold, demonstrating that the constructs in the model are distinct and not significantly overlapping.

5.4. Model fit, explanatory power, and predictive ability

R^2 is used to evaluate the explanatory power of the model, indicating the proportion of variance in the dependent variable that can be explained by the independent variables [120]. As shown in Table 6, the R^2 values for Anxiety and Self-esteem are both 0.238, suggesting that the model explains approximately 23.8 % of the variance in these constructs. The R^2 for Self-identity is 0.289, indicating a slightly higher explanatory power compared to other constructs. Additionally, all R^2 values are greater than 0, confirming that the model exhibits a

Table 4
Fornell-Larcker.

Construct	1	2	3	4	5	6	7	8
1. Anxiety	0.895							
2. Education experience	0.060	0.845						
3. Entertainment experience	0.061	0.196	0.827					
4. Escapist experience	0.148	0.058	0.196	0.843				
5. Esthetic experience	0.105	0.073	0.295	0.236	0.821			
6. Self-estrangement	0.477	0.203	0.241	0.227	0.327	0.820		
7. Self-esteem	-0.423	-0.144	-0.198	-0.114	-0.130	-0.477	0.848	
8. Virtual self-identity	0.301	0.259	0.449	0.253	0.326	0.437	-0.303	0.818

Table 5
HTMT.

Construct	1	2	3	4	5	6	7	8
1. Anxiety								
2. Education experience	0.080							
3. Entertainment experience	0.069	0.230						
4. Escapist experience	0.164	0.078	0.222					
5. Esthetic experience	0.111	0.077	0.337	0.269				
6. Self-estrangement	0.531	0.231	0.279	0.253	0.367			
7. Self-esteem	0.448	0.157	0.218	0.123	0.133	0.516		
8. Virtual self-identity	0.339	0.304	0.535	0.294	0.357	0.507	0.332	

Table 6
R² and Q² values for model constructs.

	R ²	AdjustedR ²	Q ²
Anxiety	0.238	0.234	0.185
Self-estrangement	0.191	0.189	0.126
Self-esteem	0.238	0.234	0.168
Virtual self-identity	0.289	0.281	0.183

reasonable level of explanatory capability.

Q² is used to assess the predictive relevance of the model, determining whether the model provides meaningful incremental value when predicting new data. A higher Q² value indicates stronger predictive relevance for a given variable [121]. Based on Table 6, the results demonstrate that the research model exhibits good predictive relevance.

Standardized Root Mean Square Residual (SRMR) is an indicator used to assess model residuals, with values closer to 0 indicating a better fit. Typically, an SRMR value below 0.08 is considered indicative of a well-fitting model [122]. In this study, the SRMR value for the saturated model is 0.050, and the SRMR value for the estimated model is 0.066, both of which are below the 0.08 threshold, as shown in Table 7. These results suggest that the model demonstrates a good fit.

5.5. Path coefficient analysis

The path coefficients for the measured relationships are presented in Table 8. The analysis shows that entertainment experience positively influences self-identity ($\beta = 0.335$, $t = 6.767$, $p < 0.05$), educational experience positively influences self-identity ($\beta = 0.172$, $t = 3.977$, $p < 0.05$), esthetic experience positively influences self-identity ($\beta = 0.183$, $t = 4.044$, $p < 0.05$), and escapist experience positively influences self-identity ($\beta = 0.135$, $t = 2.601$, $p < 0.05$). These results indicate that H1, H2, H3, and H4 are supported. Furthermore, self-identity positively influences anxiety ($\beta = 0.114$, $t = 2.034$, $p < 0.05$), supporting H5. Self-

Table 8
Path coefficients.

H	Paths	β	S.D.	T	P	Result
H1	Entertainment experience-> Virtual self-identity	0.335	0.049	6.767	0.000	Supported
H2	Education experience-> Virtual self-identity	0.172	0.043	3.977	0.000	Supported
H3	Esthetic experience-> Virtual self-identity	0.183	0.045	4.044	0.000	Supported
H4	Escapist experience-> Virtual self-identity	0.135	0.052	2.601	0.009	Supported
H5	Virtual self-identity-> Anxiety	0.114	0.056	2.034	0.042	Supported
H6	Virtual self-identity-> Self-estrangement	0.437	0.048	9.057	0.000	Supported
H7	Virtual self-identity-> Self-esteem	-0.117	0.053	2.195	0.028	Supported
H8	Virtual self-identity-> Self-estrangement-> Anxiety	0.187	0.032	5.904	0.000	Supported
H9	Virtual self-identity-> Self-estrangement-> Self-esteem	-0.186	0.030	6.285	0.000	Supported

identity positively influences self-estrangement ($\beta = 0.437$, $t = 9.057$, $p < 0.05$), supporting H6. Self-identity negatively influences self-esteem ($\beta = -0.117$, $t = 2.195$, $p < 0.05$), supporting H7. Self-identity indirectly influences anxiety through self-estrangement ($\beta = 0.187$, $t = 5.904$, $p < 0.05$), supporting H8. Self-identity indirectly influences self-esteem through self-estrangement ($\beta = -0.186$, $t = 6.285$, $p < 0.05$), supporting H9.

The structural equation model (SEM) analysis of the relationships between variables is illustrated in Fig. 3.

6. Discussion

Drawing on the Experience Economy Theory (EET), Self-Identity Theory, and Self-Discrepancy Theory, this study explores the mechanisms through which metaverse games experiences influence users' virtual self-identity and psychological states. A theoretical model was constructed with virtual self-identity as a mediating variable, and the Structural Equation Model (SEM) was employed to verify the effects of

Table 7
SRMR Values.

	SRMR
Saturated Model	0.050
Estimated Model	0.066

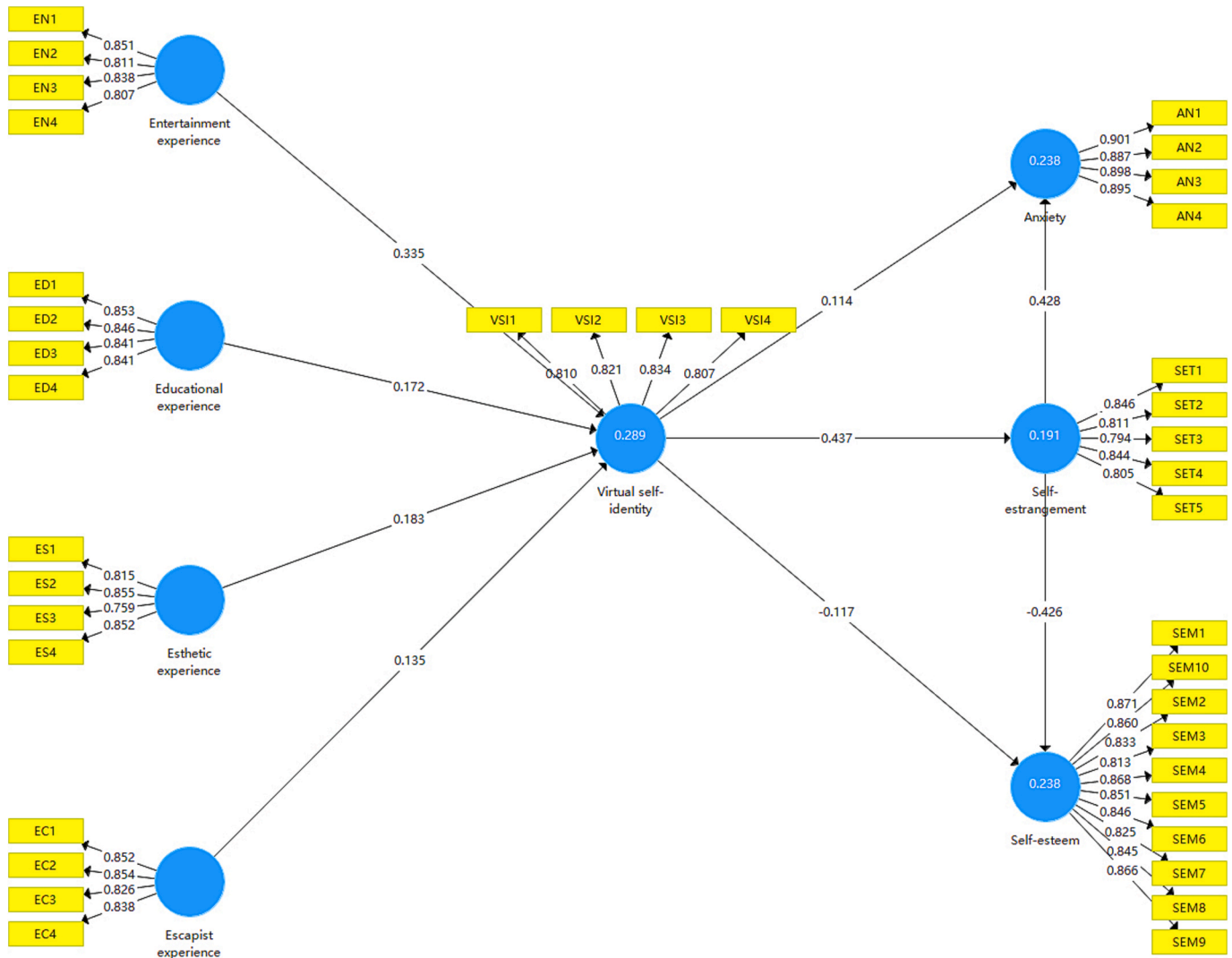


Fig. 3. Structural equation model analysis diagram.

the four dimensions of EET on virtual self-identity, as well as the effects of virtual self-identity on anxiety, self-esteem, and self-estrangement. To further understand the psychological mechanisms revealed by these paths, this section discusses the findings from three perspectives: the differential effects of experiential dimensions, the psychological effects of virtual self-identity, and the mediating role of self-estrangement.

First, the results confirm that the four dimensions of EET have significant effects on virtual self-identity, although their relative importance varies. Entertainment experience exerts the strongest influence, consistent with previous studies [123], indicating that enjoyment and interactive engagement are key sources of virtual self-identity. Esthetic and educational experiences also play important roles, aligning with the findings of Loureiro [124] and Laberger et al. [125], suggesting that high-quality visual presentation and educational content can enhance users' sense of virtual self-value and identification. In contrast, escapist experience has the weakest effect, which differs not only from the initial expectation but also from some existing studies [49,123]. This difference may be related to the fact that participants in this study primarily engaged in PC- or mobile-based metaverse games. In such non-VR environments, users are more likely to remain connected with the real world, which reduces the intensity of immersive "escape-from-reality" experiences.

Second, the study finds that virtual self-identity has significant but opposite effects on users' psychological states. Virtual self-identity

shows a significant positive effect on anxiety, consistent with previous findings. Prior studies have suggested that when users strongly identify with their virtual selves, the psychological gap between the virtual and real self may lead to dissatisfaction or stress, thereby increasing anxiety [7,56]. Meanwhile, virtual self-identity has a significant negative effect on self-esteem, which also aligns with earlier results [89,95], indicating that stronger virtual identification may weaken individuals' positive evaluation of their real selves. These findings reveal the dual psychological effects of virtual self-identity in metaverse games: on the one hand, constructing a virtual avatar allows users to experience a sense of achievement; on the other hand, when a gap arises between the idealized avatar and real-world capability, this strengthened virtual identification can negatively influence real-life emotions, resulting in increased anxiety and lower self-esteem.

Finally, this study verified the key role of self-estrangement in the relationship between virtual self-identity and psychological outcomes. The results show that self-estrangement significantly strengthens the indirect effects of virtual self-identity on anxiety and self-esteem. This finding indicates that a stronger virtual self-identity is associated with a higher level of psychological estrangement, reflecting that individuals may experience a certain sense of separation between their virtual and real selves. Compared with existing studies, this research provides empirical support for the role of self-estrangement in linking virtual experiences with real-world psychological states and further explains

how virtual experiences influence real psychological conditions through the process of identity formation.

In summary, this study integrated the perspectives of Experience Economy Theory and virtual self-identity to construct and validate a model of how metaverse games experiences affect users' psychological states. The results show that different types of experiences vary in their effects on virtual self-identity, and virtual self-identity plays a key role in bridging virtual and real psychological domains, further influencing users' anxiety and self-esteem through self-estrangement. Overall, the findings deepen the understanding of the relationship between game experiences and users' psychological responses in metaverse environments and provide empirical evidence for future research on virtual identity construction and its impact on real-world psychological well-being.

7. Conclusion

7.1. Theoretical contributions

First, this study applied the Experience Economy Theory (EET) to the field of metaverse games user behavior and verified its applicability. The results indicate that the four experiential dimensions of entertainment, education, esthetic, and escapism all have significant effects on virtual self-identity. This suggests that the EET framework can effectively explain the mechanisms through which multidimensional experiences in virtual environments shape users' psychological engagement, thereby extending the theoretical application of EET.

Second, this study supports and expands the applicability of Self-identity Theory and Self-Discrepancy Theory in virtual contexts. The results show that virtual self-identity not only directly affects anxiety and self-esteem but also indirectly influences individuals' psychological states through the psychological mechanism of self-estrangement. This finding reveals the psychological transformation pathway within the process of virtual identity construction and provides new theoretical evidence for understanding how virtual identification is associated with real psychological outcomes.

Overall, by integrating the Experience Economy Theory, Self-Identity Theory, and Self-Discrepancy Theory, this study developed an analytical framework that includes the experience layer, internalization layer, and response layer. This framework systematically explains the mechanism linking experience, identity, and psychology, offering theoretical support and an integrative perspective for future research on metaverse games user experiences and psychological processes.

7.2. Practical contributions

The findings of this study provide practical implications for the design of metaverse games and the management of users' psychological well-being. The positive transformation from the experience layer to the internalization layer helps enhance users' virtual self-identity and psychological engagement. However, the transformation from the internalization layer to the response layer may be accompanied by negative psychological outcomes such as anxiety, decreased self-esteem, and self-estrangement. Therefore, maintaining psychological balance in both design and management is an important issue for metaverse games development.

First, several suggestions are proposed for metaverse games developers. At the design level, developers should balance the effects of different experiential dimensions. The results indicate that entertainment, esthetic, and educational experiences significantly strengthen users' virtual self-identity, whereas escapist experience has a relatively weaker effect. Moderate enhancement of immersion can increase user engagement, but excessive emphasis on escapist experience may lead to detachment from reality. Developers can improve esthetic and educational experiences through high-quality visual presentation, character modeling, and task mechanisms while incorporating elements

connected to the real world, such as AR integration or real-world reward systems, to establish a healthy balance between immersion and reality.

At the psychological level, the results suggest that developers and platforms should pay attention to the potential psychological impacts of virtual self-identity. While a strong virtual identification can enhance engagement, long-term reinforcement of an idealized self may trigger anxiety and self-estrangement. To mitigate these risks, platforms may introduce psychological adjustment mechanisms, such as reflective feedback systems that help users recognize the gap between the virtual and real self, or social and cooperative functions that strengthen emotional connections to reality, thereby reducing the psychological burden caused by excessive immersion.

Second, for metaverse games users, the study reveals that virtual self-identity can bring psychological satisfaction but may also induce anxiety and self-estrangement. When engaging in metaverse games, users should rationally perceive the achievements and sense of identity brought by their virtual avatars while maintaining self-reflection and connection with reality. Actively participating in educational or esthetic experiences, as well as social and cooperative activities, can help enhance positive emotions and emotional bonding, reducing isolation and psychological disparity.

Finally, the psychological risks identified in this study, including anxiety, lower self-esteem, and self-estrangement, provide references for policymakers and industry regulators. Future governance of metaverse games ecosystems should place greater emphasis on user mental health and ethical standards to promote the sustainable and healthy development of these platforms.

7.3. Limitations and future research directions

This study has several limitations that should be addressed in future research.

First, as the data were collected through a cross-sectional survey, it is difficult to capture the dynamic relationships among variables over time. Future studies could adopt longitudinal research designs to examine how virtual self-identity and psychological states evolve across different time points or contexts, thereby providing a more comprehensive understanding of causal paths and underlying mechanisms.

Second, regarding the measurement of virtual self-identity, this study mainly focused on users' emotional attachment to and self-projection onto their virtual characters. Although this operationalization has been widely adopted in previous literature, it may not fully capture the multidimensional and dynamic nature of virtual self-identity. Future research could incorporate multidimensional scales, behavioral data, or qualitative approaches to gain a deeper understanding of the evolution of virtual self-identity.

Third, this study used a combination of convenience and purposive sampling to recruit respondents through Chinese social media platforms. Although measures such as multiple rounds of invitations, anonymity assurances, and the inclusion of low-frequency users were employed to reduce bias, the sample may still lack full representativeness. As the metaverse gaming user base continues to expand, future research could employ probability sampling or cross-cultural comparative studies to enhance the generalizability and external validity of the findings.

Moreover, this study primarily focused on metaverse games accessible through mobile or computer platforms, without including high-immersion environments that require VR or AR devices. Future studies could further compare users' psychological and behavioral differences across varying levels of immersion to enrich the understanding of how immersive experiences influence user perceptions and responses.

Finally, this study examined the relationships among virtual self-identity, anxiety, self-esteem, and self-estrangement, without considering other potential psychological variables. Subsequent studies could incorporate additional psychological or social factors to build a more comprehensive theoretical model.

Overall, future research could further refine research design,

measurement tools, sample diversity, and variable scope to deepen the understanding of metaverse games users' psychological mechanisms and provide stronger theoretical and practical foundations.

Funding statement

This work is supported by research projects of Qingdao University of Science and Technology (WST2021020) and Kookmin University.

CRediT authorship contribution statement

HanXi Li: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **YuanXiong Liu:** Writing – original draft, Software, Methodology, Investigation, Data curation. **Younghwan Pan:** Validation, Resources, Project administration, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to express their gratitude to all participants who contributed to this study, as well as to colleagues and reviewers for their valuable insights and constructive feedback that helped improve this research.

Data availability

Data will be made available on request.

References

- [1] S.Y. Mohammed, M. Aljanabi, T.R. Gadekallu, Navigating the Nexus: a systematic review of the symbiotic relationship between the metaverse and gaming, *Int. J. Cognit. Comput. Eng.* 5 (2024) 88–103, <https://doi.org/10.1016/j.ijcce.2024.02.001>.
- [2] Z. Allam, A. Sharifi, S.E. Bibri, D.S. Jones, J. Krogstie, The metaverse as a virtual form of smart cities: opportunities and challenges for environmental, economic, and social sustainability in urban futures, *smart, Cities* 5 (2022) 771–801, <https://doi.org/10.3390/smartcities5030040>.
- [3] M.M. Rezapour, A. Fatemi, M.A. Nematbakhsh, A methodology for using players' chat content for dynamic difficulty adjustment in metaverse multiplayer games, *Appl. Soft Comput.* 156 (2024) 111497, <https://doi.org/10.1016/j.asoc.2024.111497>.
- [4] N. Ali, Employing metaverse marketing through gaming and its impact on customer experience: evidence from multiple online gaming platforms, *Qas* 24 (2023), <https://doi.org/10.47750/QAS/24.196.25>.
- [5] J. Lee, T.S. Lee, S. Lee, J. Jang, S. Yoo, Y. Choi, Y.R. Park, Development and application of a metaverse-based social skills training program for children with autism spectrum disorder to improve social interaction: protocol for a randomized controlled trial, *JMIR Res. Protocols* 11 (2022) e35960, <https://doi.org/10.2196/35960>.
- [6] GrandViewResearch, Metaverse In Gaming Market Size & Share Report, 2030, 2023, <https://www.grandviewresearch.com/industry-analysis/metaverse-gaming-market-report> (accessed November 11, 2024).
- [7] M. Saker, J. Frith, Contiguous identities: the virtual self in the supposed Metaverse, *First Monday* (2022), <https://doi.org/10.5210/fm.v27i3.12471>.
- [8] R. Fraser, J. Slatery, I. Yakovenko, Escaping through video games: using your avatar to find meaning in life, *Comput. Hum. Behav.* 144 (2023) 107756, <https://doi.org/10.1016/j.chb.2023.107756>.
- [9] P. Cheremnykh, Gaming the Metaverse: user experience analysis and insights from Fortnite, 2024, https://www.theseus.fi/bitstream/handle/10024/850495/Cheremnykh_Pedcharat.pdf?sequence=2 (accessed November 11, 2024).
- [10] H. Jo, J.K. Lee, A comprehensive analysis of presence, seamlessness, and player responses in metaverse gaming, *Comput. Hum. Behav.* 159 (2024) 108345, <https://doi.org/10.1016/j.chb.2024.108345>.
- [11] Y. Seo, R. Dolan, M. Buchanan-Oliver, Playing games: advancing research on online and mobile gaming consumption, *Internet Res.* 29 (2019) 289–292, <https://doi.org/10.1108/INTR-04-2019-542>.
- [12] S. Sun, J.H. Kim, K.M. Lee, D. Nan, Exploring the association between the Proteus effect and intention to play massive multiplayer online role-playing games (MMORPGs), *Internet Res.* 34 (2024) 58–78, <https://doi.org/10.1108/INTR-07-2022-0487>.
- [13] N.J. Thomas, R. Baral, O.S. Crocco, S. Mohanan, A framework for gamification in the metaverse era: how designers envision gameful experience, *Technol. Forecast. Soc. Chang.* 193 (2023) 122544, <https://doi.org/10.1016/j.techfore.2023.122544>.
- [14] H. Oh, A.M. Fiore, M. Jeoung, Measuring experience economy concepts: tourism applications, *J. Travel Res.* 46 (2007) 119–132, <https://doi.org/10.1177/0047287507304039>.
- [15] U. Chakraborty, S.K. Biswal, Diffusion of innovation in direct-to-avatar (D2A): a study of immersive communication and branding in the metaverse, *Comput. Hum. Behav.* 158 (2024) 108318, <https://doi.org/10.1016/j.chb.2024.108318>.
- [16] S.R. Sioni, M.H. Burleson, D.A. Bekerian, Internet gaming disorder: Social phobia and identifying with your virtual self, *Comput. Hum. Behav.* 71 (2017) 11–15, <https://doi.org/10.1016/j.chb.2017.01.044>.
- [17] C. Kim, S.-G. Lee, M. Kang, I became an attractive person in the virtual world: users' identification with virtual communities and avatars, *Comput. Hum. Behav.* 28 (2012) 1663–1669, <https://doi.org/10.1016/j.chb.2012.04.004>.
- [18] T.L. Burleigh, V. Stavropoulos, L.W.L. Liew, B.L.M. Adams, M.D. Griffiths, Depression, internet gaming disorder, and the moderating effect of the gamer-avatar relationship: an exploratory longitudinal study, *Int. J. Ment. Health Addict.* 16 (2018) 102–124, <https://doi.org/10.1007/s11469-017-9806-3>.
- [19] N. Kang, D. Ding, M.B. Van Riemsdijk, N. Morina, M.A. Neerinx, W.-P. Brinkman, Self-identification with a virtual experience and its Moderating effect on self-efficacy and presence, *Int. J. Human-Comput. Interact.* 37 (2021) 181–196, <https://doi.org/10.1080/10447318.2020.1812909>.
- [20] E. Klinger, S. Bouchard, P. Légeron, S. Roy, F. Lauer, I. Chemin, P. Nugues, Virtual reality therapy versus cognitive behavior therapy for social phobia: a preliminary controlled study, *Cyberpsychol. Behav.* 8 (2005) 76–88, <https://doi.org/10.1089/cpb.2005.8.76>.
- [21] E. Park, Examining metaverse game platform adoption: Insights from innovation, behavior, and coolness, *Technol. Soc.* 77 (2024) 102594, <https://doi.org/10.1016/j.techsoc.2024.102594>.
- [22] P. Mirza-Babaei, R. Robinson, R. Mandryk, J. Pirker, C. Kang, A. Fletcher, Games and the Metaverse, in: Extended abstracts of the 2022 annual symposium on computer-human interaction in play, association for computing machinery, New York, NY, USA, 2022, pp. 318–319. DOI: 10.1145/3505270.3558355.
- [23] B.K. Wiederhold, Metaverse games: game changer for healthcare? *Cyberpsychol. Behav. Soc. Netw.* 25 (2022) 267–269, <https://doi.org/10.1089/cyber.2022.29246.editorial>.
- [24] B. Alma Çalli, Ç. Ediz, Top concerns of user experiences in Metaverse games: a text-mining based approach, *Entertain. Comput.* 46 (2023) 100576, <https://doi.org/10.1016/j.entcom.2023.100576>.
- [25] B. Gao, H. Xie, Y. Wang, Z. Xie, B. Gao, H. Xie, Y. Wang, Z. Xie, Principles of environmental and social psychology in UI/UX design for metaverse social games: a case study of Horizon Worlds, *Environ. Soc. Psychol.* 9 (2023), <https://doi.org/10.54517/esp.v9i1.1871>.
- [26] H. Jo, S. Park, J. Jeong, J. Yeon, J.K. Lee, Metaverse gaming: analyzing the impact of self-expression, achievement, social interaction, violence, and difficulty, *Behav. Inform. Technol.* (2024) 1–15, <https://doi.org/10.1080/0144929X.2024.2341059>.
- [27] C. Boo, A. Suh, Developing scales for assessing metaverse characteristics and testing their utility, *Comput. Hum. Behav. Rep.* 13 (2024) 100366, <https://doi.org/10.1016/j.chbr.2023.100366>.
- [28] D. Volk, Co-creative game development in a participatory Metaverse, 2008.
- [29] H.J. Oh, J. Kim, J.J.C. Chang, N. Park, S. Lee, Social benefits of living in the metaverse: the relationships among social presence, supportive interaction, social self-efficacy, and feelings of loneliness, *Comput. Hum. Behav.* 139 (2023) 107498, <https://doi.org/10.1016/j.chb.2022.107498>.
- [30] A.Z.H. Yee, J.R.H. Sng, Animal crossing and COVID-19: a qualitative study examining how video games satisfy basic psychological needs during the pandemic, *Front. Psychol.* 13 (2022) 800683, <https://doi.org/10.3389/fpsyg.2022.800683>.
- [31] Jeremy winslow, Minecraft Reached 140 Million Monthly Users And Generated Over \$350 Million To Date, *GameSpot* (2021). <https://www.gamespot.com/articles/minecraft-reached-140-million-monthly-users-and-generated-over-350-million-to-date/1100-6490962/> (accessed November 11, 2024).
- [32] S.-M. Park, Y.-G. Kim, A. Metaverse, Taxonomy, components, applications, and open challenges, *IEEE Access* 10 (2022) 4209–4251, <https://doi.org/10.1109/ACCESS.2021.3140175>.
- [33] J. Lin, Embodiment of User Generated Content in Electronic Game: A Case Study of the Party Game Eggy Party, 2024.
- [34] P. Swami, Metaverse: Transforming the User Experience in the Gaming and Entertainment Industry, in: Research, Innovation, and Industry Impacts of the Metaverse, IGI Global, 2024, pp. 115–128, doi: 10.4018/979-8-3693-2607-7.ch007.
- [35] A. Taylor, The Metaverse and user wellbeing: a psychological perspective, masters_research, Manchester Metropolitan University, 2024. <https://e-space.mmu.ac.uk/636485/> (accessed November 11, 2024).
- [36] E.-J. Lee, W. Lee, I. Bae, What is the draw of the metaverse? Personality correlates of zepeto use motives and their associations with psychological well-being, *Cyberpsychol. Behav. Soc. Network.* 26 (2023) 161–168. DOI: 10.1089/cyber.2022.0164.
- [37] A. Suh, How users cognitively appraise and emotionally experience the metaverse: focusing on social virtual reality, *ITP* 37 (2024) 1613–1641, <https://doi.org/10.1108/ITP-06-2022-0461>.
- [38] A.P. Zumrut, Evaluation of the Metaverse Universe in Light of Psychology and Sociology, in: F.S. Esen, H. Tinmaz, M. Singh (Eds.), Metaverse: Technologies,

- Opportunities and Threats, Springer Nature, Singapore, 2023: pp. 203–218. DOI: 10.1007/978-981-99-4641-9_14.
- [39] S.S. Usmani, M. Sharath, M. Mehendale, Future of mental health in the metaverse, *General Psychiat.* 35 (2022) e100825, <https://doi.org/10.1136/gpsych-2022-100825>.
- [40] B.J. Pine, J.H. Gilmore, Welcome to the experience economy, *Harv. Bus. Rev.* 76 (1998) 97–105.
- [41] B.J. Knutson, J.A. Beck, S. Kim, J. Cha, Service quality as a component of the hospitality experience: proposal of a conceptual model and framework for research, *J. Foodserv. Bus. Res.* 13 (2010) 15–23, <https://doi.org/10.1080/15378021003595889>.
- [42] I.K.W. Lai, Y. Liu, D. Lu, The effects of tourists' destination culinary experience on electronic word-of-mouth generation intention: the experience economy theory, *Asia Pacific J. Tour. Res.* 26 (2021) 231–244, <https://doi.org/10.1080/10941665.2020.1851273>.
- [43] Song, A study of festival visitors' loyalty based on experience economy: The case of Boryeong Mud festival, (2011). https://scholar.google.com/scholar_lookup?hl=en&volume=25&publication_year=2011&pages=179-198&journal=Korean+Journal+of+Tourism+Research&issue=6&author=H.+K.+Song&author=Y.+J.+Choi&author=C.+K.+Lee&title=A+study+of+festival+visitors%E2%80%99+loyalty+based+on+experience+economy%3A+The+case+of+Boryeong+mud+festival (accessed November 12, 2024).
- [44] Q. Zhang, X. Liu, Z. Li, Z. Tan, Multi-experiences in the art performance tourism: integrating experience economy model with flow theory, *J. Travel Tour. Mark.* 38 (2021) 491–510, <https://doi.org/10.1080/10548408.2021.1952148>.
- [45] H. Lee, T.H. Jung, M.C. tom Dieck, N. Chung, Experiencing immersive virtual reality in museums, *Informat. Manage.* 57 (2020) 103229. DOI: 10.1016/j.im.2019.103229.
- [46] M.C. tom Dieck, T.H. Jung, P.A. Rauschnabel, Determining visitor engagement through augmented reality at science festivals: An experience economy perspective, *Comput. Human Behav.* 82 (2018) 44–53. DOI: 10.1016/j.chb.2017.12.043.
- [47] S. Hosany, M. Witham, Dimensions of cruisers' experiences, satisfaction, and intention to recommend, *J. Travel Res.* 49 (2010) 351–364, <https://doi.org/10.1177/0047287509346859>.
- [48] A. Manthiou, S. (Ally) Lee, L. (Rebecca) Tang, L. Chiang, The experience economy approach to festival marketing: vivid memory and attendee loyalty, *J. Serv. Market.* 28 (2014) 22–35. DOI: 10.1108/JSM-06-2012-0105.
- [49] H.J. Song, C.-K. Lee, J.A. Park, Y.H. Hwang, Y. Reisinger, The influence of tourist experience on perceived value and satisfaction with temple stays: the experience economy theory, *J. Travel Tour. Mark.* 32 (2015) 401–415, <https://doi.org/10.1080/10548408.2014.898606>.
- [50] H. Rahman, S.A. Wahid, F. Ahmad, N. Ali, Game-based learning in metaverse: Virtual chemistry classroom for chemical bonding for remote education, *Educ. Inf. Technol.* (2024), <https://doi.org/10.1007/s10639-024-12575-5>.
- [51] F. Xiang, J. Liu, Research on interactive visual design from the perspective of metaverse, 4 (2022).
- [52] D.J. Terry, M.A. Hogg, K.M. White, The theory of planned behaviour: Self-identity, social identity and group norms, *Br. J. Soc. Psychol.* 38 (1999) 225–244, <https://doi.org/10.1348/014466699164149>.
- [53] M. Jeannerod, E. Pacherie, Agency, simulation and self-identification, *Mind Lang.* 19 (2004) 113–146, <https://doi.org/10.1111/j.1468-0017.2004.00251.x>.
- [54] Y. Lee, J. Lee, Z. Lee, Social influence on technology acceptance behavior: self-identity theory perspective, *SIGMIS Database* 37 (2006) 60–75, <https://doi.org/10.1145/1161345.1161355>.
- [55] R.W. Belk, Extended self in a digital world: table 1, *J. Consum. Res.* 40 (2013) 477–500, <https://doi.org/10.1086/671052>.
- [56] D. Costa Pinto, G. Reale, R. Segabinazzi, C.A. Vargas Rossi, Online identity construction: how gamers redefine their identity in experiential communities, *J. Consum. Behav.* 14 (2015) 399–409, <https://doi.org/10.1002/cb.1556>.
- [57] C.R. Hollenbeck, A.M. Kaikati, Consumers' use of brands to reflect their actual and ideal selves on Facebook, *Int. J. Res. Mark.* 29 (2012) 395–405, <https://doi.org/10.1016/j.jresmar.2012.06.002>.
- [58] Schau, Gilly, We Are What We Post? Self-Presentation in Personal Web Space, (2003). https://scholar.google.com/scholar_lookup?title=We%20are%20what%20we%20post%20Self-presentation%20in%20personal%20web%20space&publication_year=2003&author=J.H.%20Schau&author=M.C.%20Gilly (accessed November 13, 2024).
- [59] N. Yee, J.N. Bailenson, N. Ducheneaut, The proteus effect: implications of transformed digital self-representation on online and offline behavior, *Commun. Res.* 36 (2009) 285–312, <https://doi.org/10.1177/0093650208330254>.
- [60] A.K. Przybylski, N. Weinstein, K. Murayama, M.F. Lynch, R.M. Ryan, The ideal self at play: the appeal of video games that let you be all you can be, *Psychol. Sci.* 23 (2012) 69–76, <https://doi.org/10.1177/0956797611418676>.
- [61] T. Mancini, C. Imperato, F. Sibilla, Does avatar's character and emotional bond expose to gaming addiction? Two studies on virtual self-discrepancy, avatar identification and gaming addiction in massively multiplayer online role-playing game players, *Comput. Human Behav.* 92 (2019) 297–305. DOI: 10.1016/j.chb.2018.11.007.
- [62] J. Van Looy, C. Courtis, M. De Vocht, Player identification in online games: validation of a scale for measuring identification in MMORPGs, in: Proceedings of the 3rd International Conference on Fun and Games, ACM, Leuven Belgium, 2010: pp. 126–134. DOI: 10.1145/1823818.1823832.
- [63] E. Abdelraouf, H. Amin, Virtual facades: exploring the relationship between self-presentation on facebook and social anxiety among Egyptian generation Z users, (2024), doi: 10.1386/jamr.00079.1.
- [64] J.M. Cheek, Identity Orientations and Self-Interpretation, in: D.M. Buss, N. Cantor (Eds.), *Personality Psychology: Recent Trends and Emerging Directions*, Springer US, New York, NY, 1989: pp. 275–285. DOI: 10.1007/978-1-4684-0634-4_21.
- [65] B. Koles, P. Nagy, Virtual customers behind avatars: the relationship between virtual identity and virtual consumption in second life, *J. Theor. Appl. Electron. Commer. Res.* 7 (2012) 17–18, <https://doi.org/10.4067/S0718-18762012000200009>.
- [66] D.D. Li, A.K. Liao, A. Khoo, Player-avatar identification in video gaming: concept and measurement, *Comput. Hum. Behav.* 29 (2013) 257–263, <https://doi.org/10.1016/j.chb.2012.09.002>.
- [67] E.T. Higgins, Self-discrepancy: a theory relating self and affect, *Psychol. Rev.* 94 (1987) 319–340, <https://doi.org/10.1037/0033-295X.94.3.319>.
- [68] T.J. Strauman, E.T. Higgins, Self-Discrepancies as predictors of vulnerability to distinct syndromes of chronic emotional distress - Strauman - - Journal of Personality - Wiley Online Library, (1988). <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-6494.1988.tb00472.x> (accessed November 14, 2024).
- [69] J.-M. Lee, K.-S. Shin, A.-Y. Suh, The influence of self-discrepancy in virtual and cross worlds on individuals' activities in online communities, *J. Soc. E-Business Studies* 16 (2011) 23–45, <https://doi.org/10.7838/jsebs.2011.16.3.023>.
- [70] Y. Li, Y. Li, G. Castaño, The mechanism underlying the effect of actual-ideal self-discrepancy on internet gaming addiction: a moderated mediation model, *Int. J. Ment. Health Addict.* 19 (2021) 283–301, <https://doi.org/10.1007/s11469-020-00273-5>.
- [71] S.-A.-A. Jin, The virtual malleable self and the virtual identity discrepancy model: investigative frameworks for virtual possible selves and others in avatar-based identity construction and social interaction, *Comput. Hum. Behav.* 28 (2012) 2160–2168, <https://doi.org/10.1016/j.chb.2012.06.022>.
- [72] K. Bessière, A.F. Seay, S. Kiesler, The ideal elf identity exploration in world of Warcraft, *Cyberpsychol. Behav.* 10 (2007) 530–535, <https://doi.org/10.1089/cpb.2007.9994>.
- [73] N. Ducheneaut, M.-H. Wen, N. Yee, G. Wadley, Body and mind: a study of avatar personalization in three virtual worlds, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, Association for Computing Machinery, New York, NY, USA, 2009, pp. 1151–1160. DOI: 10.1145/1518701.1518877.
- [74] C. Hu, R. Cao, J. Huang, Y. Wei, The effect of self-discrepancy on online behavior: a literature review, *Front. Psychol.* 13 (2022), <https://doi.org/10.3389/fpsyg.2022.883736>.
- [75] H.-K. Kim, K.E. Davis, Toward a comprehensive theory of problematic internet use: evaluating the role of self-esteem, anxiety, flow, and the self-rated importance of internet activities, *Comput. Hum. Behav.* 25 (2009) 490–500, <https://doi.org/10.1016/j.chb.2008.11.001>.
- [76] C.-C. Wang, Y.Y.-H. Yang, I. Shen, Self-present by avatars in multiplayer online role-playing games: the influence of self-esteem, online disinhibition, and self-discrepancy, 2014.
- [77] T.D. Parsons, A.A. Rizzo, Affective outcomes of virtual reality exposure therapy for anxiety and specific phobias: a meta-analysis, *J. Behav. Ther. Exp. Psychiatry* 39 (2008) 250–261, <https://doi.org/10.1016/j.jbtep.2007.07.007>.
- [78] M. Alsoud, A. Trawnih, H. Yaseen, T. Majali, A.R. Alsoud, O.A. Jaber, How could entertainment content marketing affect intention to use the metaverse? Empirical findings, *Int. J. Informat. Manage. Data Insights* 4 (2024) 100258, <https://doi.org/10.1016/j.jimide.2024.100258>.
- [79] E.-S.-H. Tan, Entertainment is emotion: the functional architecture of the entertainment experience, *Media Psychol.* (2008), <https://doi.org/10.1080/15213260701853161>.
- [80] J. Chen, Y. Wu, Would you be willing to purchase virtual gifts during esports live streams? Streamer characteristics and cultural traits, *Comput. Human Behav.* 152 (2024) 108075, <https://doi.org/10.1016/j.chb.2023.108075>.
- [81] C. Scorolli, E. Naddi Grasso, L. Stacchio, V. Armandi, G. Matteucci, G. Marfia, Would you rather come to a tango concert in theater or in VR? Aesthetic emotions & social presence in musical experiences, either live, 2D or 3D, *Computers in Human Behavior* 149 (2023) 107910. DOI: 10.1016/j.chb.2023.107910.
- [82] K.R. Lodewyk, P.H. Winne, Relations among the structure of learning tasks, achievement, and changes in self-efficacy in secondary students, *J. Educ. Psychol.* 97 (2005) 3–12, <https://doi.org/10.1037/0022-0663.97.1.3>.
- [83] T. Jung, M.C. tom Dieck, H. Lee, N. Chung, Effects of Virtual Reality and Augmented Reality on Visitor Experiences in Museum, in: A. Inversini, R. Schegg (Eds.), *Information and Communication Technologies in Tourism 2016*, Springer International Publishing, Cham, 2016: pp. 621–635. DOI: 10.1007/978-3-319-28231-2_45.
- [84] H. Shupeng, Z. Huanfang, F. a. N. Cuiying, Z. Zongkui, The impacts of virtual avatar in video game on users' self-concept, *Adv. Psychol. Sci.* 28 (2020) 810. DOI: 10.3724/SP.J.1042.2020.00810.
- [85] J. Deleuze, P. Maurage, A. Schimmenti, F. Nuyens, A. Melzer, J. Billieux, Escaping reality through videogames is linked to an implicit preference for virtual over real-life stimuli, *J. Affect. Disord.* 245 (2019) 1024–1031, <https://doi.org/10.1016/j.jad.2018.11.078>.
- [86] F. Melodia, N. Canale, M.D. Griffiths, The role of avoidance coping and escape motives in problematic online gaming: a systematic literature review, *Int J Ment Health Addiction* 20 (2022) 996–1022, <https://doi.org/10.1007/s11469-020-00422-w>.
- [87] D. Pal, C. Arpanikanondt, The sweet escape to metaverse: exploring escapism, anxiety, and virtual place attachment, *Comput. Hum. Behav.* 150 (2024) 107998, <https://doi.org/10.1016/j.chb.2023.107998>.
- [88] E. Wallace, I. Buil, Antecedents and consequences of conspicuous green behavior on social media: Incorporating the virtual self-identity into the theory of planned

- behavior, *J. Bus. Res.* 157 (2023) 113549, <https://doi.org/10.1016/j.jbusres.2022.113549>.
- [89] Y. Liu, Impact of virtual identities on social media on women's self-esteem, *SHS Web of Conf.* 180 (2023) 02005, <https://doi.org/10.1051/shsconf/202318002005>.
- [90] D.J. Kuss, J. Louws, R.W. Wiers, Online gaming addiction? Motives predict addictive play behavior in massively multiplayer online role-playing games, *Cyberpsychol. Behav. Soc. Network.* 15 (2012) 480–485. DOI: 10.1089/cyber.2012.0034.
- [91] J. Van Looy, C. Courtois, M. De Vocht, L. De Marez, Player identification in online games: validation of a scale for measuring identification in MMOGs, *Media Psychol.* 15 (2012) 197–221, <https://doi.org/10.1080/15213269.2012.674917>.
- [92] D. Li, A. Liao, A. Khoo, Examining the influence of actual-ideal self-discrepancies, depression, and escapism, on pathological gaming among massively multiplayer online adolescent gamers, *Cyberpsychol. Behav. Soc. Netw.* 14 (2011) 535–539, <https://doi.org/10.1089/cyber.2010.0463>.
- [93] C.-Y. Chen, S.-L. Chang, Moderating effects of information-oriented versus escapism-oriented motivations on the relationship between psychological well-being and problematic use of video game live-streaming services, 2019, DOI: 10.1556/2006.8.2019.34.
- [94] K. Bailey, R. West, J. Kuffel, What would my avatar do? Gaming, pathology, and risky decision making, *Front. Psychol.* 4 (2013), <https://doi.org/10.3389/fpsyg.2013.00609>.
- [95] A. Fakhrou, M.A. Moussa, Virtual Self-identity and its Relationship to Self-esteem among University Students during Corona Pandemic, *Journal of Educational Sciences – Qatar University* 24 (2024). DOI: 10.29117/jes.2024.0163.
- [96] M. Seeman, On the meaning of alienation, *Am. Sociol. Rev.* 24 (1959) 783–791, <https://doi.org/10.2307/2088565>.
- [97] T. Szanto, Emotional self-alienation, *Midwest Stud Philos* 41 (2017) 260–286, <https://doi.org/10.1111/misp.12074>.
- [98] M.V. Manannikova, Alienation: identity crisis, *I.I.I.* (2023) 143–150. DOI: 10.25198/2077-7175-2023-6-143.
- [99] M. Daiiani, Avatar versus self the correlation of Iranian digital role-players with self-estrangement and identity crisis, in: 2018 2nd National and 1st International Digital Games Research Conference: Trends, Technologies, and Applications (DGRC), 2018: pp. 95–103. DOI: 10.1109/DGRC.2018.8712059.
- [100] C.J. Couch, Self-identification and alienation, *Sociol. Q.* 7 (1966) 255–264, <https://doi.org/10.1111/j.1533-8525.1966.tb01692.x>.
- [101] H. Kasinath, Interactive effect of self-actualization, self-esteem and self-alienation on academic achievement, *Artha J. Soc. Sci.* 5 (2006) 1–16, <https://doi.org/10.12724/ajss.8.1>.
- [102] L. Aymerich-Franch, R.F. Kizilcec, J.N. Bailenson, The relationship between virtual self similarity and social anxiety, *Front. Hum. Neurosci.* 8 (2014), <https://doi.org/10.3389/fnhum.2014.00944>.
- [103] J.-Y. Yen, C.-F. Yen, C.-S. Chen, P.-W. Wang, Y.-H. Chang, C.-H. Ko, Social anxiety in online and real-life interaction and their associated factors, *Cyberpsychol. Behav. Soc. Netw.* 15 (2012) 7–12, <https://doi.org/10.1089/cyber.2011.0015>.
- [104] Berg N., Non-Response Bias, (2005). <https://papers.ssrn.com/abstract=1691967> (accessed November 23, 2024).
- [105] J.R. Smith, D.J. Terry, A.S.R. Manstead, W.R. Louis, D. Kotterman, J. Wolfs, Interaction effects in the theory of planned behavior: the interplay of self-identity and past behavior, *J. Appl. Soc. Psychol.* 37 (2007) 2726–2750, <https://doi.org/10.1111/j.1559-1816.2007.00278.x>.
- [106] M. Hamilton, The Assessment of anxiety states by rating, *Br. J. Med. Psychol.* 32 (1959) 50–55, <https://doi.org/10.1111/j.2044-8341.1959.tb00467.x>.
- [107] M. Rosenberg, Rosenberg self-esteem scale, *APA PsycTests* (1965).
- [108] Soper, Free A-priori Sample Size Calculator for Structural Equation Models - Free Statistics Calculators, (2024). <https://www.danielsoper.com/statcalc/calculator.aspx?id=89> (accessed October 7, 2024).
- [109] J.F. Hair, M. Sarstedt, C.M. Ringle, J.A. Mena, An assessment of the use of partial least squares structural equation modeling in marketing research, *J. Acad. Mark. Sci.* 40 (2012) 414–433, <https://doi.org/10.1007/s11747-011-0261-6>.
- [110] P.M. Podsakoff, S.B. MacKenzie, J.-Y. Lee, N.P. Podsakoff, Common method biases in behavioral research: a critical review of the literature and recommended remedies, *J. Appl. Psychol.* 88 (2003) 879–903, <https://doi.org/10.1037/0021-9010.88.5.879>.
- [111] N. Shrestha, Detecting multicollinearity in regression analysis, *AJAMS* 8 (2020) 39–42, <https://doi.org/10.12691/ajams-8-2-1>.
- [112] N. Kock, Common method bias in PLS-SEM, *Int. J. E-Collaboration* 11 (2015), <https://doi.org/10.4018/ijec.2015100101>.
- [113] M. Sarstedt, C.M. Ringle, J.F. Hair, Partial Least Squares Structural Equation Modeling, in: C. Homburg, M. Klarmann, A. Vomberg (Eds.), *Handbook of Market Research*, Springer International Publishing, Cham, 2017: pp. 1–40. DOI: 10.1007/978-3-319-05542-8_15-1.
- [114] H. Cengiz, A. Pouyan, H. Azdemir, Linking gamers' competitive spirit and in-game impulse purchase: the need for popularity as a mediator and social competence as a moderator, *Comput. Hum. Behav.* 163 (2025) 108479, <https://doi.org/10.1016/j.chb.2024.108479>.
- [115] E. Park, S. Baek, J. Ohm, H.J. Chang, Determinants of player acceptance of mobile social network games: an application of extended technology acceptance model, *Telematics Inform.* 31 (2014) 3–15, <https://doi.org/10.1016/j.tele.2013.07.001>.
- [116] W.K. Tan, P.-H. Chiu, Theory of consumption value: a lens to examine the use and continual use intention of online game subscription services, *Comput. Hum. Behav.* 160 (2024) 108377, <https://doi.org/10.1016/j.chb.2024.108377>.
- [117] A. Christmann, S. Van Aelst, Robust estimation of Cronbach's alpha, *J. Multivar. Anal.* 97 (2006) 1660–1674, <https://doi.org/10.1016/j.jmva.2005.05.012>.
- [118] C. Fornell, D.F. Larcker, Evaluating structural equation models with unobservable variables and measurement error, *J. Mark. Res.* 18 (1981) 39, <https://doi.org/10.2307/3151312>.
- [119] J.F. Hair, J.J. Risher, M. Sarstedt, C.M. Ringle, When to use and how to report the results of PLS-SEM, *EBR* 31 (2019) 2–24, <https://doi.org/10.1108/EBR-11-2018-0203>.
- [120] G. Shmueli, O. Koppius, P. Analytics, *Inf. Syst. Res.* (2011).
- [121] W.W. Chin, Bootstrap Cross-Validation Indices for PLS Path Model Assessment, in: V. Esposito Vinzi, W.W. Chin, J. Henseler, H. Wang (Eds.), *Handbook of Partial Least Squares*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2010: pp. 83–97. DOI: 10.1007/978-3-540-32827-8_4.
- [122] J. Henseler, C.M. Ringle, M. Sarstedt, A new criterion for assessing discriminant validity in variance-based structural equation modeling, *J. of the Acad. Mark. Sci.* 43 (2015) 115–135, <https://doi.org/10.1007/s11747-014-0403-8>.
- [123] J. Hwang, S.O. Lyu, The antecedents and consequences of well-being perception: an application of the experience economy to golf tournament tourists, *J. Destin. Mark. Manage.* 4 (2015) 248–257, <https://doi.org/10.1016/j.jdmm.2015.09.002>.
- [124] S.M.C. Loureiro, The role of the rural tourism experience economy in place attachment and behavioral intentions, *Int. J. Hosp. Manag.* 40 (2014) 1–9, <https://doi.org/10.1016/j.ijhm.2014.02.010>.
- [125] M. Laberge, A. Tartarin, S.K. Lee, Exploring consumer satisfaction with an AR beauty app based on the technology acceptance model and experience economy theory, *J. Market Anal.* (2024), <https://doi.org/10.1057/s41270-024-00360-9>.